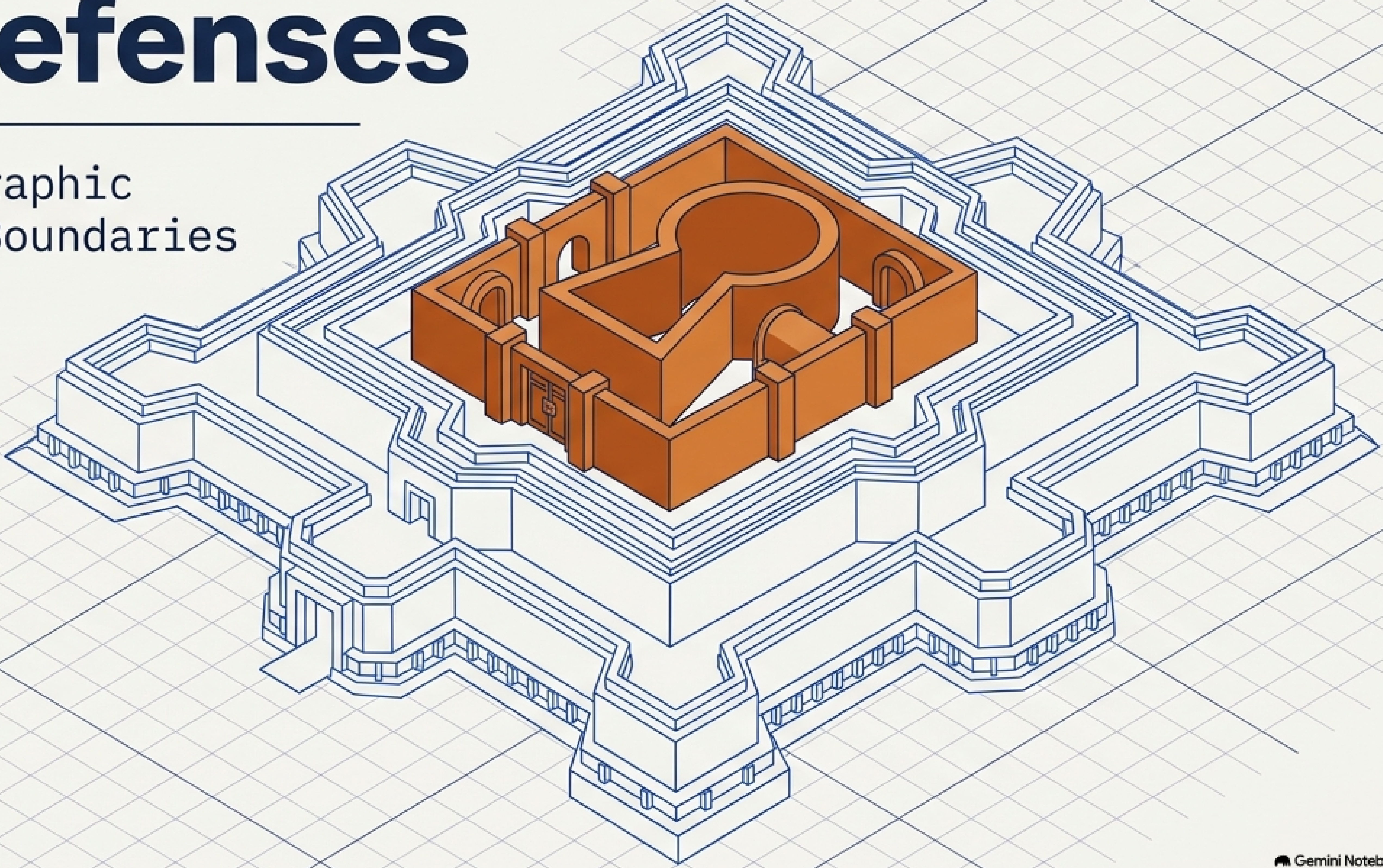
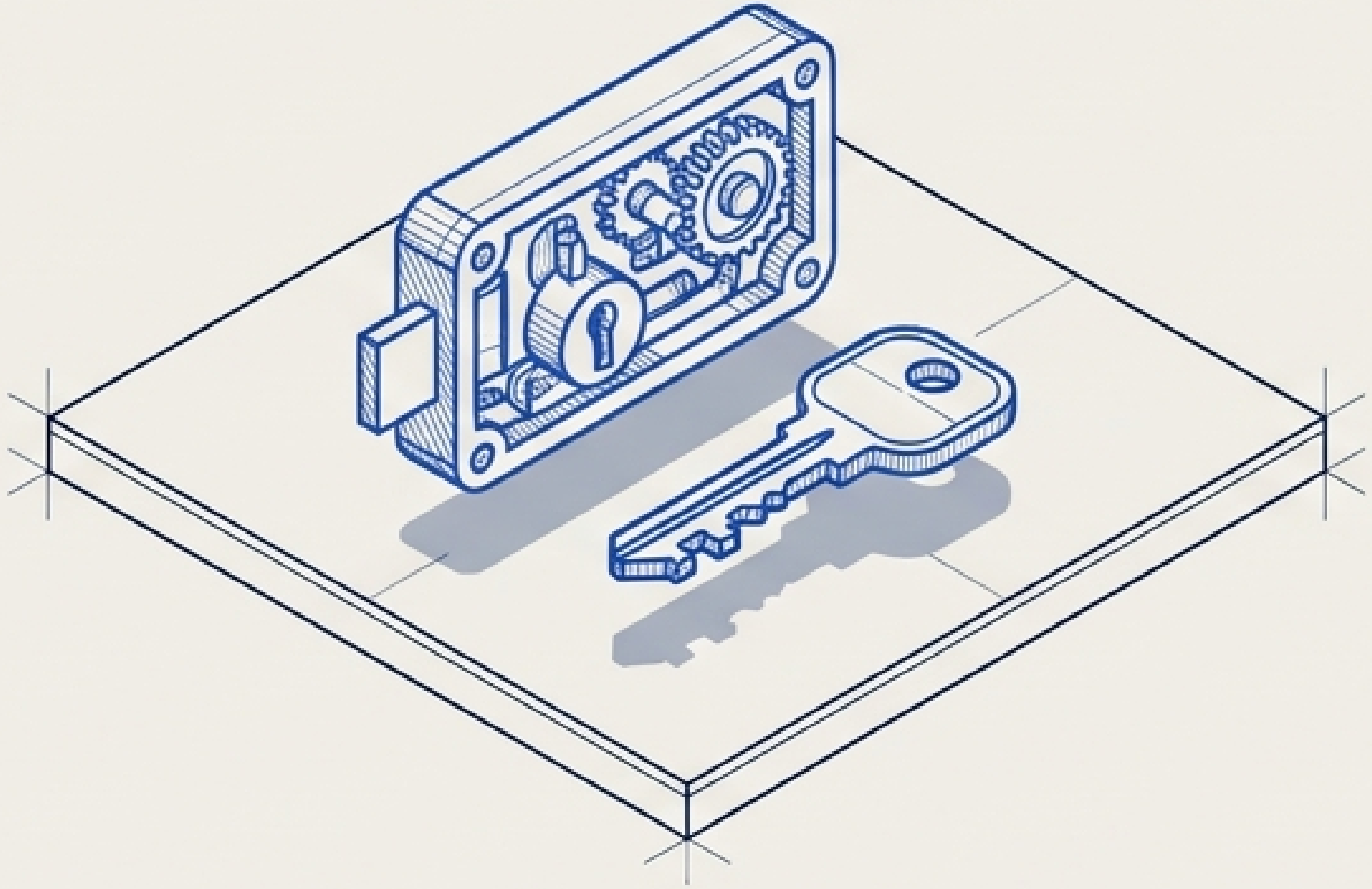
Blueprint for AWS Cloud Defenses

Mastering Cryptographic Keys and Network Boundaries



The Dual Pillars of Cloud Defense

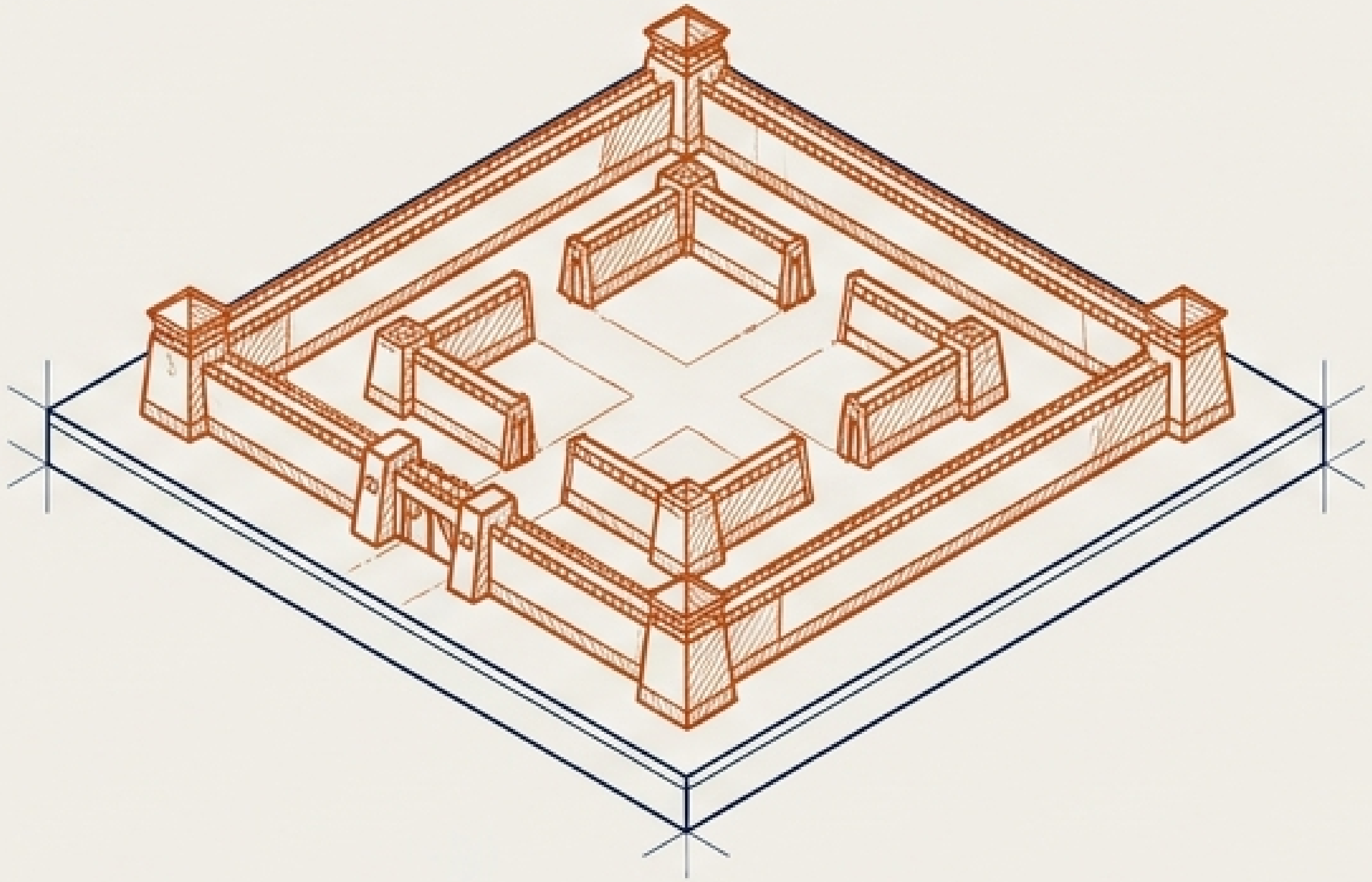
Pillar 1: Access



Who gets in?

AWS Key Pairs. The cryptographic mechanism used to securely authenticate and control access to specific server instances.

Pillar 2: Architecture



Where can they go?

AWS Subnets. The network isolation boundaries that dictate spatial architecture and traffic routing.

The Cryptographic Lock for EC2 Instances

An **AWS Key Pair** is a set of cryptographic keys used to securely authenticate and access an EC2 instance without a standard password.

Linux Systems

Provides secure, passwordless SSH access to Linux instances. Protects entirely against password-guessing attacks.

Windows Systems

Used for RDP-related access, specifically required to retrieve and decrypt the initial Windows administrator password.

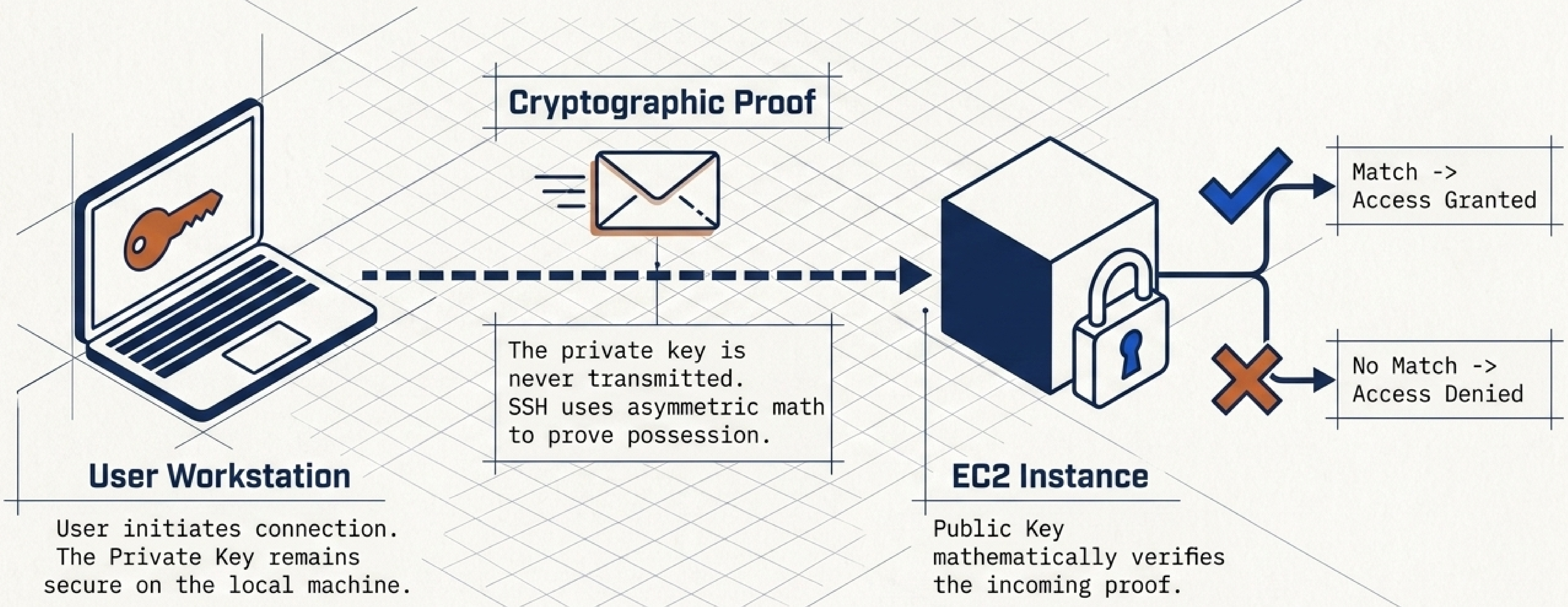
Automation & Scripting

Establishes a strong, standard authentication framework that securely supports large-scale automation and

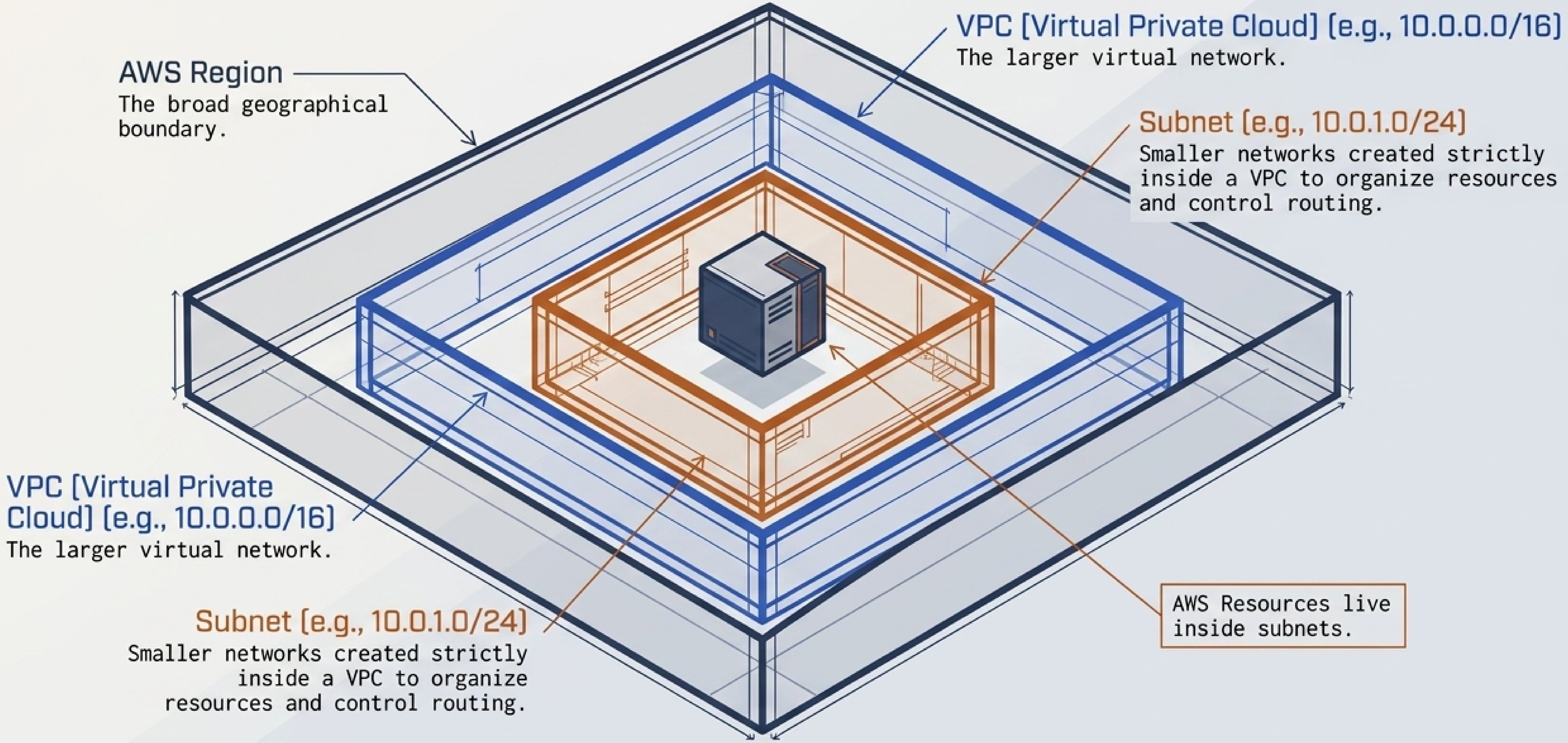
The Cryptographic Matrix: Public vs. Private

	Public Key	Private Key
Storage Location	Stored on or associated with the EC2 instance.	Downloaded to the user's local machine upon creation.
Sharing Rules	Safe to share publicly.	**Must be kept absolute secret. Never commit to source control.**
AWS Visibility	AWS retains and uses this to verify the user.	**AWS does not need it and does not retain a copy.** Losing it blocks access permanently.
Primary Role	Verifies the user's identity.	Proves the user's identity.

The Authentication Circuit



Establishing the Network Perimeter

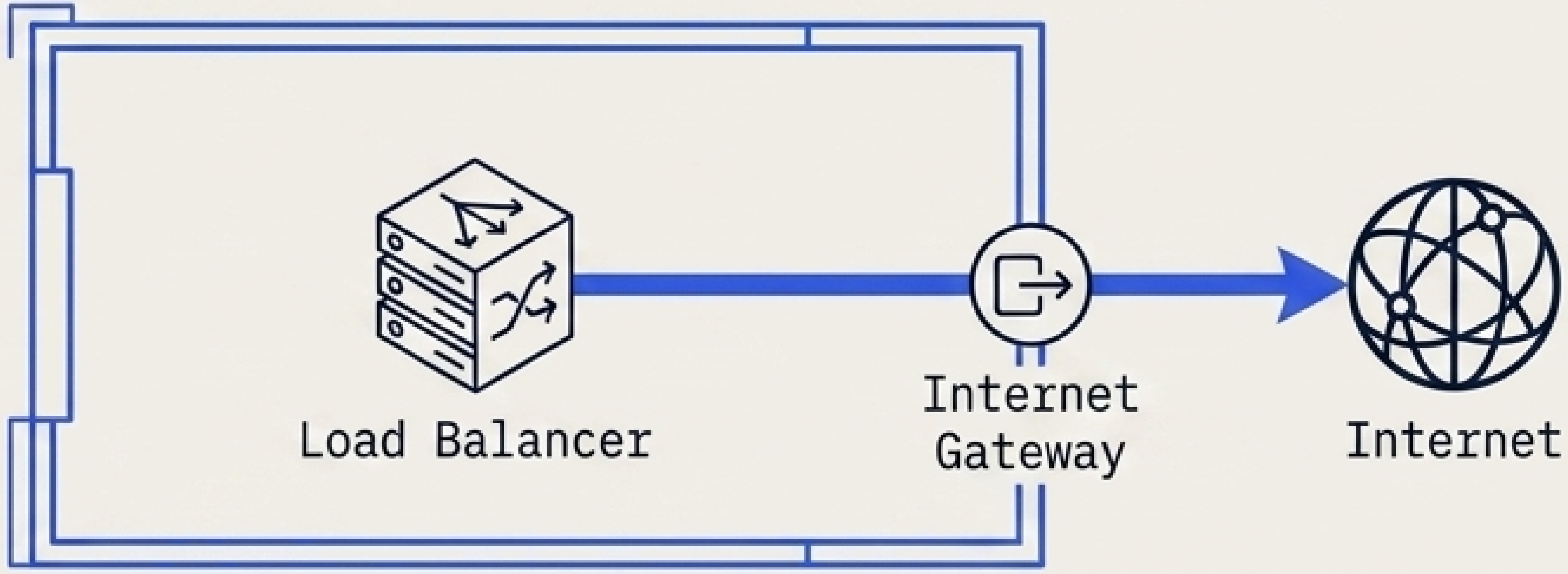


Routing Architecture: Public vs. Private Subnets

Public Subnet

Definition: Has a direct routing path to the Internet Gateway.

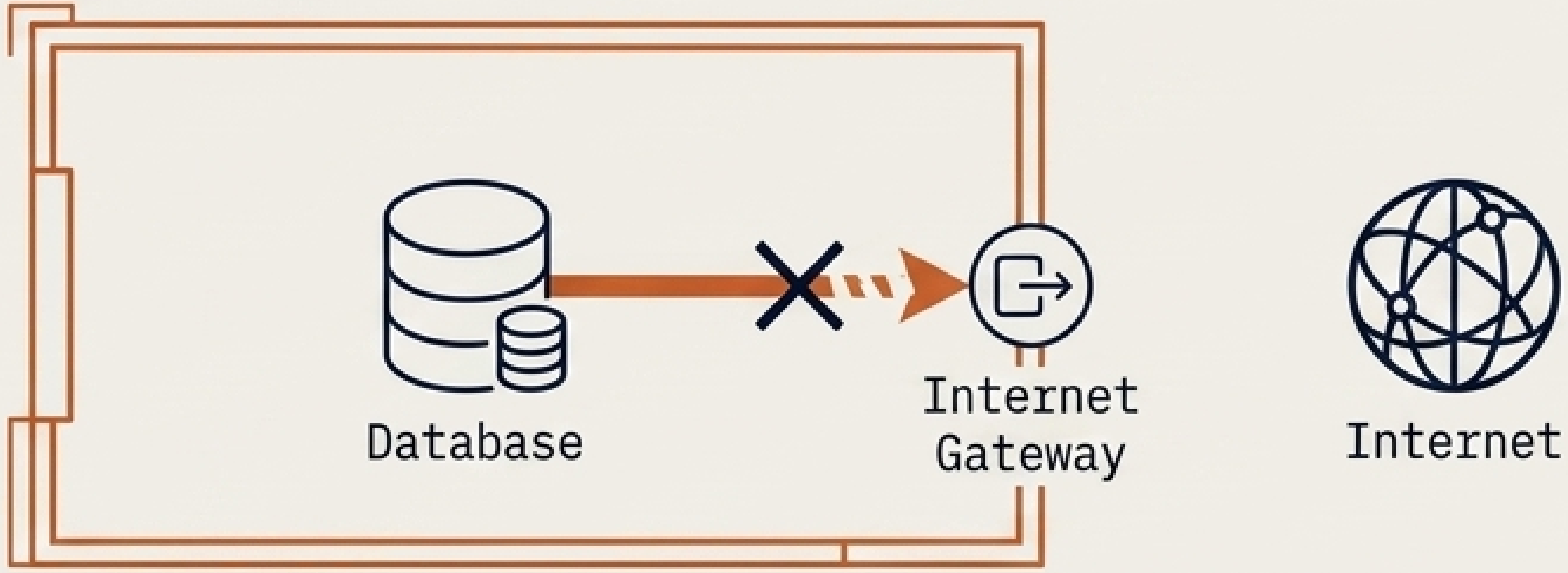
Use Cases: Internet-facing load balancers, public application components.



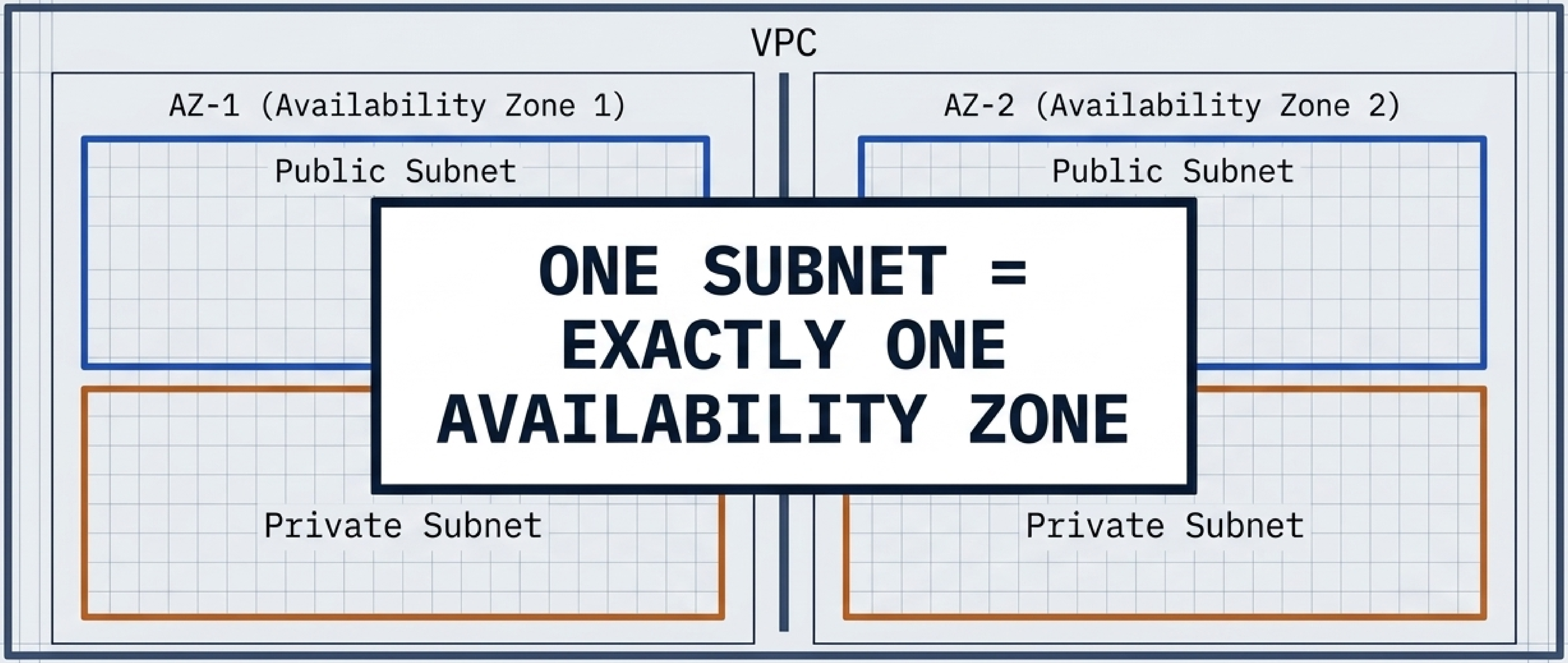
Private Subnet

Definition: No direct route to the Internet Gateway.

Use Cases: Backend application servers, internal services, databases. (Can use a NAT Gateway for outbound-only traffic).

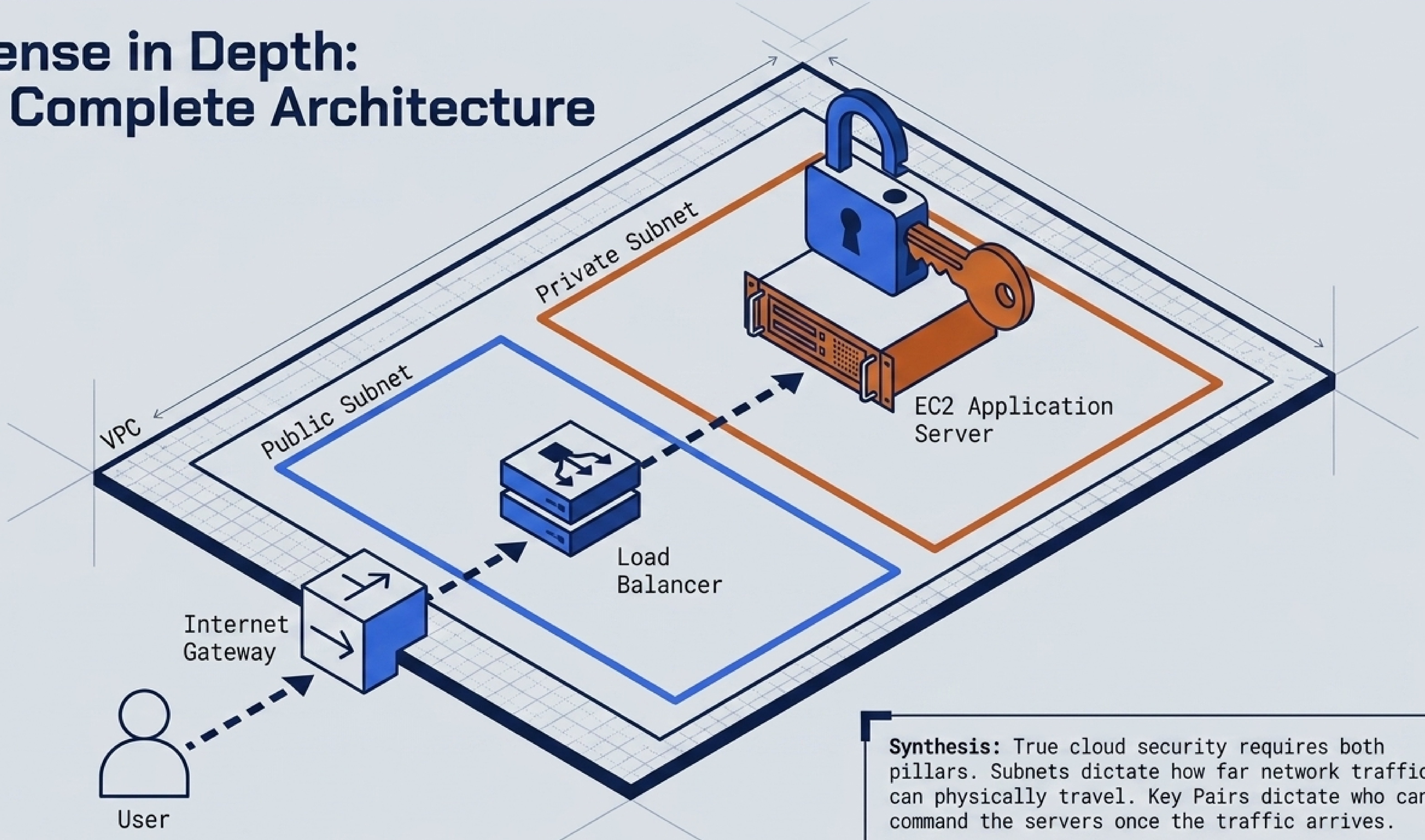


The Golden Rule of High Availability



While a single subnet cannot span multiple physical boundaries, a VPC contains subnets distributed across multiple AZs to prevent a single data center failure from causing an outage.

Defense in Depth: The Complete Architecture



Synthesis: True cloud security requires both pillars. Subnets dictate how far network traffic can physically travel. Key Pairs dictate who can command the servers once the traffic arrives.

Immutable Axioms of Cloud Infrastructure

AXIOM I: NEVER share your Private Key. AWS does not have a copy to give back to you.

AXIOM II: The Public Key verifies identity; the Private Key mathematically proves it.

AXIOM III: A Subnet is bound to exactly ONE Availability Zone. It cannot cross physical boundaries.

AXIOM IV: The ONLY difference between a Public and Private subnet is a configured routing path to an Internet Gateway.